

PICK AND PLACE MACHINE

Selection Made Easy

If you are planning to purchase a new pick and place machine for your manufacturing unit, here are a few insights from experts that could help you select the best machine for your specific requirement

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A pick and place (PnP) machine makes use of comprehensive mechanisms to mount small electronic components and SMDs, such as capacitors, resistors, and ICs, onto a printed circuit board (PCB). It ensures that the small components are picked up carefully and placed accurately onto the PCB. The machine can be manual requiring placement by hand, semi-automatic with software assistance for placing, or automatic for typical mass manufacturing units with in-line facilities. Combined with their respective strengths, here are a few parameters to consider before investing in the machine.

The PnP machines have considerably improved recently to achieve speed with accuracy, and flexibility. However, standard requirements for specific parameters, like accuracy rate, component sizes, and board sizes, have not changed much. The future of

SMT production is anything but predictable according to experts. With rapidly changing build schedules, there is a proliferation of new materials and advanced components to locate, load, mount, and trace. As these high-mix challenges go mainstream and mid-volume, real productivity and throughput suffer. Several equipment manufacturing firms are trying to solve precisely these challenges.

Let us have a look at some of the features and the challenges so that you can select the most appropriate machine for your production house.

Machine construction. When selecting a PnP machine, the most important factor is its construction. It determines the machine's effective components per hour (CPH) number, footprint, and the number of component feeders it can accommodate. The construction may either be all-welded steel or a bolted-together frame.

The most accurate machine may come with a welded structure to provide significant stability for accurate positioning and high-speed movement in X and Y axes. This suits any production environment without need for calibration. Apart from this, careful selection of centering methods and component positioning systems can support optimisation.

Placement speed. CPH is the measure at which the machine will pick up and place components onto the PCB. The speed measured in CPH will have a direct impact on production volume. Many PnP machine manufacturers rate their machines based on the IPC 9850 standard, a general guide for part mix and PCB placement arrays of their machines. It helps determine the throughput after accounting for feeder changes, transfer



Yamaha Motor's Z:TA-R YSM40R (Credit: <https://global.yamaha-motor.com>)